9th Class • Punjab Board • PCTB New Syllabus

Chapter 8: Bioenergetics

Solved notes in book order — MCQs, short questions, long questions, inquisitive questions and key points.

Chapter 8 • Units 8.1 – 8.3 • Exercise solved in book order

What this chapter is about

• Bioenergetics is the study of how living things obtain, change, store and use energy. Cells keep a ready supply of usable energy in ATP.
• Photosynthesis stores light energy in glucose. Cellular respiration releases the energy of food and transfers it to ATP.
• This chapter explains ATP, light and dark reactions of photosynthesis, aerobic respiration, fermentation and the uses of respiratory energy.
• Every exercise question is answered below in the same order as in the textbook, with clear equations and comparison tables.

A. Multiple Choice Questions

Select the correct option. A short reason is given after every answer.

1. When we get energy from ATP, which bonds are broken?

a) P–P bonds b) C–H bonds
c) C–N bonds d) C–O bonds

Answer: a) P–P bonds — ATP releases usable energy when the high-energy bond between its phosphate groups is hydrolysed and ADP is formed.

2. Light reactions of photosynthesis occur in;

a) Plasma membrane of cell b) Cytoplasm of cell
c) Stroma of chloroplasts d) Thylakoids of chloroplasts

Answer: d) Thylakoids of chloroplasts — Chlorophyll and the electron carriers of the light reactions are present in thylakoid membranes.

3. Which type of chlorophyll is most common in plants?

a) Chlorophyll a b) Chlorophyll b
c) Chlorophyll c d) Chlorophyll d

Answer: a) Chlorophyll a — Chlorophyll-a is the main photosynthetic pigment; chlorophyll-b and carotenoids act as accessory pigments.

4. Which wavelengths of light are absorbed to maximum by chlorophylls?

a) Green and blue b) Green and red
c) Red and blue d) Only green

Answer: c) Red and blue — Chlorophyll absorbs red and blue light strongly, while most green light is reflected or transmitted.

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5. When yeast ferments glucose, the products are;

a) Alcohol and CO_2 b) Alcohol and water
c) Lactic acid d) CO_2 and H_2O

Answer: a) Alcohol and CO_2 — Yeast carries out alcoholic fermentation and forms ethanol and carbon dioxide without oxygen.

6. Where do the dark reactions of photosynthesis occur?

a) Stroma of chloroplast b) Thylakoids of chloroplast
c) Outer membrane d) Cytoplasm

Answer: a) Stroma of chloroplast — The Calvin cycle uses ATP and NADPH in the stroma to reduce carbon dioxide and make carbohydrate.

7. Which molecule donates electrons in the light-dependent reactions of photosynthesis?

a) NADPH b) Water
c) Oxygen d) Carbon dioxide

Answer: b) Water — Water splits during photolysis; its hydrogen atoms provide electrons, and oxygen is released.

8. Which process in aerobic respiration produces the most ATP?

a) Glycolysis b) Krebs cycle
c) Electron transport chain d) Fermentation

Answer: c) Electron transport chain — Energy released by electrons passing through the chain is used to make most of the cell's ATP.

9. How many ATP molecules are the net profit from one glucose molecule during anaerobic respiration?

a) 2 b) 4
c) 12 d) 36

Answer: a) 2 — Only glycolysis supplies a net gain of two ATP; fermentation itself adds no further ATP.

10. What is a common byproduct of anaerobic respiration in animal cells?

a) Oxygen b) Water
c) Lactic acid d) Carbon dioxide

Answer: c) Lactic acid — When oxygen is insufficient, animal cells such as skeletal muscle cells convert pyruvic acid into lactic acid.

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B. Write Short Answers

1. Write the importance of oxidation-reduction reactions.

Oxidation-reduction (redox) reactions transfer electrons between molecules. Oxidation is the loss of electrons and reduction is the gain of electrons. The flow of electrons releases or transfers energy, so these reactions are essential for making ATP during cellular respiration and photosynthesis.

2. What do ATP and ADP mean? What are the roles of these molecules for the cellular metabolism?

ATP means **adenosine triphosphate**. It is the cell's energy currency. Its hydrolysis to ADP and inorganic phosphate supplies energy for movement, active transport, synthesis of biomolecules and other cell activities.

ADP means **adenosine diphosphate**. It is the lower-energy, two-phosphate form. Energy released from food joins ADP with inorganic phosphate (P_i) to recharge it into ATP. In this way ATP and ADP transfer energy between energy-releasing and energy-using reactions.

$$ADP + P_i + \text{energy} \rightleftharpoons ATP + H_2O$$

3. Write down the word equation for photosynthesis.

Chlorophyll is needed to capture the light energy.

Carbon dioxide + water + light energy → glucose + oxygen + water

4. Why is chlorophyll important for photosynthesis?

Chlorophyll absorbs sunlight, mainly blue and red wavelengths. The absorbed light excites its electrons, and their energy is used to form ATP and NADPH. These two energy-rich molecules then drive the formation of glucose in the dark reactions.

5. How is oxygen produced during photosynthesis?

During the light reactions, light energy splits water molecules. This is called photolysis. Hydrogen provides electrons and hydrogen ions, while oxygen is released as a by-product through the stomata.

6. Which organisms carry out photosynthesis? Which cell organelle is responsible for the absorption of light for photosynthesis?

Plants, algae and some bacteria carry out photosynthesis. In plants and algae, the **chloroplast** is the organelle responsible; its thylakoid membranes contain chlorophyll and absorb light.

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7. State the main purpose of cellular respiration?

The main purpose of cellular respiration is to oxidise food, usually glucose, and transfer its stored energy to ATP. Cells then use this ATP for growth, movement, repair, transport and other life processes.

8. State the equation (in words or symbols) for aerobic respiration.

Glucose + oxygen → carbon dioxide + water + energy (ATP)

$$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy (ATP)}$$

9. Write a brief note on the role of oxygen in aerobic respiration.

Oxygen is the final electron acceptor in the electron transport chain. It accepts electrons and hydrogen ions and forms water. This allows the complete oxidation of glucose and the release of a large amount of ATP.

10. Define anaerobic and aerobic respiration.

Anaerobic respiration	Aerobic respiration
It occurs **without oxygen**.	It occurs **in the presence of oxygen**.
Glucose is incompletely oxidised, so less energy is released.	Glucose is completely oxidised, so much more energy is released.
It forms lactic acid in animals, or ethanol and CO_2 in yeast.	Its end-products are carbon dioxide and water, together with ATP.

11. What are the end-products of anaerobic respiration in animals and yeast?

In animals, anaerobic respiration converts pyruvic acid into **lactic acid**. In yeast, alcoholic fermentation produces **ethyl alcohol (ethanol) and carbon dioxide**. Both pathways have a net gain of only two ATP per glucose molecule.

12. How do muscles respond to oxygen deficiency during intense exercise?

During intense exercise, the oxygen supply may become insufficient for the muscles' energy demand. Skeletal muscles then respire anaerobically for a short time and make ATP, but lactic acid also collects. This causes tiredness and muscle discomfort; normal breathing later supplies oxygen to remove the oxygen debt.

13. List ways in which respiratory energy is used in the body.

Respiratory energy (ATP) is used for **muscle contraction and movement; active transport; synthesis of biomolecules; DNA replication and mitosis; transmission of nerve impulses; maintenance of body temperature; and breakdown and elimination of toxins** from the body.

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C. Write Answers in Detail

1. Explain ATP as a molecule that is the chief energy currency of all cells.

ATP (adenosine triphosphate) is a nucleotide made of **adenine** (a nitrogenous base), **ribose** (a five-carbon sugar) and **three phosphate groups**. The bonds between the phosphate groups are high-energy bonds. Fritz Lipman proposed in 1941 that ATP is the main energy-transfer molecule in cells.

When a cell needs energy, the terminal phosphate group of ATP is removed by hydrolysis. ATP becomes ADP and inorganic phosphate (P_i), and about **7.3 kcal per mole** becomes available under standard conditions. This energy is used directly for cell work.

$$ATP + H_2O \rightarrow ADP + P_i + \text{energy (7.3 kcal/mole)}$$

$$ADP + P_i + \text{energy from food} \rightarrow ATP + H_2O$$

ATP is called an **energy currency** because it links energy-releasing and energy-consuming reactions. Energy from the oxidation of food is stored by making ATP; the same ATP is later spent on muscle contraction, active transport, biosynthesis, DNA replication and nerve impulses.

ATP gives small, usable amounts of energy exactly where they are needed. It is repeatedly regenerated from ADP, so cells do not have to break down glucose every time a small amount of energy is required.

2. Outline the processes involved in photosynthesis?

Photosynthesis is the synthesis of glucose from carbon dioxide and water in the presence of sunlight and chlorophyll; oxygen is released as a by-product. It occurs in two connected phases.

$$6CO_2 + 12H_2O + \text{light energy} \rightarrow C_6H_{12}O_6 + 6O_2 + 6H_2O$$

1. Light reactions — thylakoid membranes: Chlorophyll absorbs light and its electrons gain energy. These electrons pass through an electron transport chain, and their energy helps make ATP. Light also splits water: oxygen is released, while electrons and hydrogen ions are supplied. Finally, $NADP^+$ is reduced to NADPH.

2. Dark reactions (Calvin cycle) — stroma: Carbon dioxide combines with a five-carbon compound. The unstable six-carbon product splits into two three-carbon compounds. ATP and hydrogen from NADPH reduce these compounds to three-carbon carbohydrates, which are used to make glucose. Some three-carbon molecules regenerate the original five-carbon compound, using more ATP, so the cycle continues.

Therefore, the light reactions capture light energy as ATP and NADPH, and the dark reactions use that chemical energy to fix carbon dioxide into carbohydrate. Chlorophyll-a is the main pigment; chlorophyll-b and carotenoids absorb additional wavelengths as accessory pigments.

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3. Write a note on the intake of carbon dioxide and water by plants

Intake of carbon dioxide: Carbon dioxide enters the leaf from the air by diffusion through tiny pores called **stomata**. Guard cells open and close these pores. Inside the leaf, carbon dioxide moves through air spaces and dissolves on the moist surfaces of mesophyll cells. It then enters the cells and reaches the chloroplast stroma for the Calvin cycle.

Intake of water: Root hairs provide a very large absorbing surface. Their cell sap has a higher solute concentration than soil water, so water enters the root hairs by **osmosis**. It then passes through the epidermis, cortex, endodermis and pericycle to the root xylem.

Transport to leaves: Xylem vessels carry water and dissolved mineral salts upwards to the stem and leaves. Evaporation from leaf surfaces creates a transpirational pull, while cohesion between water molecules and adhesion to xylem walls maintain a continuous water column. Water is used in the light reactions, where it supplies electrons and hydrogen ions and releases oxygen.

4. Explain the types and importance of anaerobic respiration.

Anaerobic respiration (fermentation) releases energy from glucose **without oxygen**. Glucose is only partly oxidised, so the net gain is two ATP per glucose molecule. Glycolysis is the same as in aerobic respiration, but pyruvic acid then follows one of two pathways.

1. Alcoholic fermentation: In yeast and some bacteria, pyruvic acid is converted into ethyl alcohol and carbon dioxide.

$$C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 + \text{energy (2 ATP)}$$

2. Lactic acid fermentation: In human and other animal muscles, and in some bacteria, pyruvic acid is converted into lactic acid when oxygen is not available in sufficient amount.

$$C_6H_{12}O_6 \rightarrow 2C_3H_6O_3 + \text{energy (2 ATP)}$$

Importance: Early organisms obtained energy anaerobically before free oxygen appeared on Earth. Present-day anaerobes still use this process. Human skeletal muscles use it temporarily during hard exercise when oxygen supply cannot meet demand.

Fermentation is also useful in industry. Bacterial fermentation helps make cheese and yogurt; yeast fermentation is used in brewing and bread baking; and fungal fermentation is used in making soy sauce.

5. Outline the mechanism of aerobic respiration.

Aerobic respiration completely oxidises glucose in the presence of oxygen and transfers a large amount of its energy to ATP. It has three main stages.

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1. Glycolysis — cytoplasm: One six-carbon glucose molecule breaks into two three-carbon pyruvic acid molecules. The net products are **two ATP and two NADH**. Oxygen is not needed, so glycolysis also occurs in anaerobic respiration.

2. Krebs cycle — mitochondrial matrix: When oxygen is present, pyruvic acid enters the mitochondrion. Each pyruvic acid first forms acetyl coenzyme-A, releasing CO_2 and forming NADH. Acetyl-CoA is then completely oxidised in the Krebs cycle. Carbon dioxide, ATP, NADH and $FADH_2$ are produced.

3. Electron transport chain — inner mitochondrial membrane: NADH and $FADH_2$ release high-energy electrons and hydrogen ions and change back to NAD and FAD. As electrons pass through a series of carriers, their released energy makes ATP from ADP and P_i. At the end, oxygen accepts the electrons and hydrogen ions and forms water. This stage produces most of the ATP.

Glucose + oxygen → carbon dioxide + water + energy (ATP)

6. Compare the processes of respiration and photosynthesis.

Photosynthesis	Respiration
Type: an anabolic process; glucose is built from simple substances.	**Type:** a catabolic process; glucose is broken down.
Energy: light energy is stored in the chemical bonds of food.	**Energy:** bond energy of food is transferred to ATP.
Main equation: $CO_2 + H_2O \rightarrow$ glucose + O_2.	**Main equation:** glucose + $O_2 \rightarrow CO_2 + H_2O$.
Organisms: some bacteria, all algae and all plants.	**Organisms:** all living organisms.
Site: chloroplasts, in green parts only.	**Site:** cytoplasm and mitochondria, in all living cells.
Time: only in the presence of light.	**Time:** continues all the time.
Gases: takes in CO_2 and releases O_2.	**Gases:** aerobic respiration takes in O_2 and releases CO_2.
Food: glucose is produced and may be stored.	**Food:** glucose is used to release energy.

D. Inquisitive Questions

1. How does the structure of ATP enable it to store and release energy efficiently?

ATP is small, soluble and has three phosphate groups attached to adenosine. The phosphate groups carry like negative charges, so their close arrangement stores considerable chemical potential energy. The terminal phosphate can be removed by a simple hydrolysis reaction, giving ADP and P_i and making energy quickly available.

Only a manageable amount of energy is transferred at one time, so little is wasted. ATP can move to the exact place where work is needed, and ADP can be recharged with energy from food. This rapid, reversible ATP–ADP cycle makes ATP an efficient short-term energy carrier rather than a long-term energy store like glucose or fat.

E. More Important Questions (from the text)

1. Differentiate between aerobic and anaerobic respiration.

Aerobic respiration	Anaerobic respiration
Requires oxygen.	Does not require oxygen.
Glucose is completely oxidised.	Glucose is incompletely oxidised.
Releases much more energy.	Has a net gain of only 2 ATP per glucose.
Final products are CO_2 and H_2O.	Final products are lactic acid, or ethanol and CO_2.
Main site is cytoplasm and mitochondria.	Occurs in the cytoplasm.
It is the main energy source of most organisms.	It supplies anaerobes and gives temporary energy to muscles during oxygen shortage.

2. Differentiate between light and dark reactions of photosynthesis.

Light reactions	Dark reactions (Calvin cycle)
Occur in the thylakoid membranes.	Occur in the stroma of the chloroplast.
Directly require light energy.	Do not directly need light, but depend on ATP and NADPH from light reactions.
Chlorophyll absorbs light and water is split.	Carbon dioxide is fixed and reduced to carbohydrate.
ATP and NADPH are produced; oxygen is released.	ATP and NADPH are used; glucose is formed and the five-carbon acceptor is regenerated.

Points to Remember

Bioenergetics: the study of how organisms acquire, convert, store and use energy.

Redox reactions: oxidation means loss of electrons and reduction means gain of electrons. Electron flow is important for ATP production.

ATP: adenine + ribose + three phosphates. ATP is the cell's energy currency; hydrolysis of its terminal phosphate releases about 7.3 kcal/mole under standard conditions.

Photosynthesis: carbon dioxide and water form glucose in sunlight and chlorophyll; oxygen is released.

Light reactions: occur in thylakoids and produce ATP and NADPH. Water supplies electrons and hydrogen ions, and oxygen is released.

Dark reactions: occur in the stroma. ATP and NADPH are used to reduce carbon dioxide and make carbohydrate in the Calvin cycle.

Chlorophyll-a: the main pigment, absorbing mainly blue and red light. Chlorophyll-b and carotenoids are accessory pigments.

Aerobic respiration: complete oxidation of glucose with oxygen; carbon dioxide and water are formed and much ATP is produced.

Glycolysis: occurs in the cytoplasm and forms two pyruvic acid molecules, with a net gain of two ATP and two NADH.

Krebs cycle: occurs in the mitochondrial matrix and forms CO_2, ATP, NADH and $FADH_2$.

Electron transport chain: occurs on the inner mitochondrial membrane and makes most ATP. Oxygen is the final electron acceptor and forms water.

Fermentation: anaerobic respiration with a net gain of two ATP. Animals form lactic acid; yeast forms ethanol and carbon dioxide.

How to attempt this chapter in the paper

- Learn the sites: light reactions — thylakoids; Calvin cycle — stroma; glycolysis — cytoplasm; Krebs cycle — mitochondrial matrix; electron transport chain — inner mitochondrial membrane.
- Write balanced equations with proper subscripts. In anaerobic respiration, mention the organism before writing the end-product.
- For ATP, include its three structural parts, hydrolysis to ADP + P_i, energy value and two uses.
- For comparisons, draw a two-column table and write matching points on the same row.

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